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# MTH301 Final Term Papers By Waqar (File 5)

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(t^4, \sqrt{t+1}, \frac{3}{t^2}\right)$

Answer ( Please select your correct option )

VuAnswers.com

- $\vec{r}'(t) = \left(4t^3, \frac{1}{\sqrt{t+1}}, \frac{-6}{t^3}\right)$
- $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{6}{t^3}\right)$
- $\vec{r}'(t) = \left(4t^4, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3}\right)$
- $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3}\right)$ correct

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(e^{t^2}, t^2, \sec 2t\right)$

Answer ( Please select your correct option )

VuAnswers.com

- $\vec{r}'(t) = \left(2te^{t^2}, 2t, 2\sec 2t \tan 2t\right)$
- $\vec{r}'(t) = \left(te^{t^2}, 2t, \sec 2t \tan 2t\right)$ correct
- $\vec{r}'(t) = \left(2te^{t^2}, 2t, \tan 2t\right)$
- $\vec{r}'(t) = \left(t^2e^{t^2}, 2t, \sec 2t \tan 2t\right)$

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

Evaluate the integral $\int \left[(3t-1)\hat{i} + \sqrt{t}\,\hat{j}\right] dt$

Answer ( Please select your correct option )

VuAnswers.com

- $\int \left[(3t-1)\hat{i} + \sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t^2 - t\right)\hat{i} + \left(\frac{3}{2}t^{\frac{3}{2}}\right)\hat{j} + C$
- $\int \left[(3t-1)\hat{i} + \sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t^2 - t\right)\hat{i} + \left(\frac{2}{3}t^{\frac{3}{2}}\right)\hat{j} + C$ correct
- $\int \left[(3t-1)\hat{i} + \sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t - t\right)\hat{i} + \left(\frac{2}{3}t^{\frac{3}{2}}\right)\hat{j} + C$
- $\int \left[(3t-1)\hat{i} + \sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t^2 - t\right)\hat{i} + \left(\frac{1}{2}t^{\frac{3}{2}}\right)\hat{j} + C$

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

The following differential is exact
$dz = (6xy + 2y^2 - 5)\,dx + (3x^2 + 4xy - 6)\,dy$

Answer ( Please select your correct option )

VuAnswers.com

- True — correct
- False

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the amplitude of a periodic function defined by $f(x) = 4\cos 3x$ ?

Answer ( Please select your correct option )

VuAnswers.com

- 1
- 3
- 4 — correct
- 12

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = 4\sin 2x$ ?

Answer ( Please select your correct option )

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- $2^0$
- $4^0$
- $8^0$
- $180^0$ — correct
- $360^0$

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

Answer ( Please select your correct option )

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- $\frac{\pi}{2}$
- $\pi$
- $\frac{3\pi}{2}$
- $4\pi$

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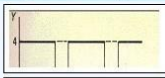
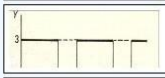
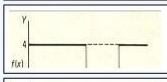
Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

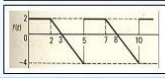
$$f(x) = \begin{cases} 4 & 0 < x < 5 \\ 0 & 5 < x < 8 \end{cases}$$

Answer ( Please select your correct option )

VuAnswers.com

- 
- 
- 

correct

- 

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

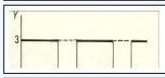
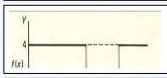
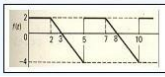
$$f(x) = \begin{cases} 4 & 0 < x < 6 \\ 0 & 6 < x < 8 \end{cases}$$

Answer ( Please select your correct option )

VuAnswers.com

- 

correct

- 
- 
- 

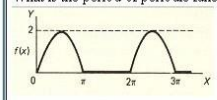
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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?




Answer ( Please select your correct option )

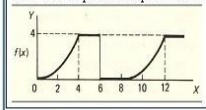
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- 0
- 2
- $\pi$
- $2\pi$ correct

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?




Answer ( Please select your correct option )

VuAnswers.com

- 0
- 4
- 6
- 8 correct

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

Let $L$ denotes the Laplace Transform.

If $L\{F(t)\} = f(s)$ where $s$ is a constant and $\lim_{t\to 0}\left(\frac{F(t)}{t}\right)$ exists then which of the following equation holds?

Answer ( Please select your correct option )

VuAnswers.com

- $L\left(\frac{F(t)}{t}\right) = f(s+a)$
- $L\left(\frac{F(t)}{t}\right) = f(s-a)$
- $L\left(\frac{F(t)}{t}\right) = \int_s^\infty f(s)\,ds$ correct
- $L\left(\frac{F(t)}{t}\right) = -\frac{d}{ds}(f(s))$

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

The function $f(x) = x^2 \cos 2x$ is ----------

Answer ( Please select your correct option )

VuAnswers.com

- Neither even nor odd
- Odd function
- Even function correct

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of an even function is symmetrical about ----------

Answer ( Please select your correct option )

VuAnswers.com

- x-axis
- y-axis correct
- origin

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of an odd function is symmetrical about ----------

Answer ( Please select your correct option )

VuAnswers.com

- x-axis
- y-axis
- origin correct

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Sign of line integral is reversed when ----------

Answer ( Please select your correct option )

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- path of integration is divided into parts.
- path of integration is parallel to y-axis.
- direction of path of integration is reversed. correct
- path of integration is parallel to x-axis.

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

What is laplace transform of the function $F(t)$ if $F(t) = \cos 2t$ ?

Answer ( Please select your correct option )

VuAnswers.com

- $L\{\cos 2t\} = \frac{2}{s^2+4}$ correct
- $L\{\cos 2t\} = \frac{1}{s-2}$
- $L\{\cos 2t\} = \frac{s}{s^2+4}$
- $L\{\cos 2t\} = \frac{2!}{s^3}$

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

What is Laplace Inverse Transform of $\frac{5}{s^2+25}$

Answer ( Please select your correct option )

VuAnswers.com

- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 5t$ correct
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 5t$
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 25t$
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 25t$

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

Answer ( Please select your correct option )

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- 1
- 0
- 2 correct
- -2

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Evaluate the line integral $\int_C (xy)\,dx+(1+y^2)\,dy$ where C is the line segment from (1, 0) to (3, 0).

Answer ( Please select your correct option )

VuAnswers.com

- -4
- 0 correct
- 4
- Do not exist

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

If $p$ is the period of a function then that function is said to be periodic if $f(x+p)=f(x)$, _____

Answer ( Please select your correct option )

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- For all values of x in the domain of f correct
- For positive values of x in the domain of f
- For negative values of x in the domain of f

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

A vector field is a vector each of whose components is a scalar field

Answer ( Please select your correct option )

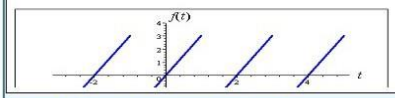
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- True
- False

correct

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)



Answer ( Please select your correct option )

VuAnswers.com

- An odd function
- An even function
- Neither even nor odd

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following set is the union of sets of rational and irrational numbers?

Answer ( Please select your correct option )

VuAnswers.com

- Set of rational numbers
- Set of integers
- Set of real numbers

correct

- Empty set.

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ shows . . . . . $(x_1, y_1, z_1)$ and $(x_2, y_2, z_2)$.

Answer ( Please select your correct option )

VuAnswers.com

- Distance between the points — correct
- Midpoint of the line joining the points
- Ratio between the points

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Domain of the function $f(x, y) = \sqrt{y - x^2}$ satisfies the condition . . . . . .

Answer ( Please select your correct option )

VuAnswers.com

- $y < x^2$
- $y \geq x^2$ — correct
- $y \neq x^2$
- Entire space

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

If rectangular co-ordinates of a point are $(1, \sqrt{3}, -2)$, then value of "r" in cylindrical co-ordinates is . . . . . .

Answer ( Please select your correct option )

VuAnswers.com

- $\sqrt{2}$
- 2
- $2\sqrt{2}$
- $-2\sqrt{2}$

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Suppose $f(x, y) = x^3 e^{xy}$. Which of the following options is correct?

Answer ( Please select your correct option )

VuAnswers.com

- $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$
- $\frac{\partial f}{\partial y} = x^3 e^{xy}$
- $\frac{\partial f}{\partial y} = x^4 e^{xy}$ correct
- $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

The function decreases most rapidly in the direction of . . . . . .

Answer ( Please select your correct option )

VuAnswers.com

- $-\nabla f$ correct
- $-|\nabla f|$
- $\nabla f \times \hat{u}$
- $|\nabla f|$

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

Two surfaces are said to be orthogonal at the point of their intersection if their normals at that point are ---------

Answer ( Please select your correct option )

VuAnswers.com

- Parallel
- Perpendicular correct
- In opposite direction
- Overlapping

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

For a function $f(x,y)$ to have both absolute maximum and minimum, it must be Continuous on . . . . . set R.

Answer ( Please select your correct option )

VuAnswers.com

- a closed and bounded
- an open and bounded
- a closed and unbounded
- an open and unbounded

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x,y) : 2 \le x \le 4 \text{ and } 0 \le y \le 1\}$, then $\iint_R (4xe^{2y})dA =$ . . . . . .

Answer ( Please select your correct option )

VuAnswers.com

- $\int_0^1 \int_2^4 (4xe^{2y})dydx$
- $\int_0^1 \int_2^4 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dydx$

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

Let R be a closed region in two dimensional space then the double integral over R calculates. . . . . .

Answer ( Please select your correct option )

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- Area of R
- Radius of inscribed circle in R.
- Distance between two endpoints of R.
- None of these

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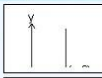
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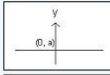
Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

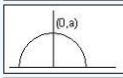
Match the following equation in polar co-ordinates with its graph.
$r \cos\theta = a$
*where a is an arbitrary constant*

Answer ( Please select your correct option )

VuAnswers.com









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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

Answer ( Please select your correct option )

VuAnswers.com

- $\left(2, \frac{-\pi}{4}\right)$
- $\left(2, \frac{-\pi}{2}\right)$
- $\left(2, \frac{-\pi}{3}\right)$
- $\left(2, \frac{3\pi}{4}\right)$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing $(r, \theta)$ by $(r, -\theta)$ then the curve is said to be symmetric about .....

Answer ( Please select your correct option )

VuAnswers.com

- Initial line
- y-axis correct
- Pole
- origin

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of curve $r^2 = 4 \cos 2\theta$ is symmetric about .....

Answer ( Please select your correct option )

VuAnswers.com

- Initial line correct
- y-axis
- both initial line and y-axis
- none of these

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

If $p(r,\theta)$ is a point in polar coordinate system, then $r$ is the distance of $p$ from .....

Answer ( Please select your correct option )

VuAnswers.com

- Pole
- Imaginary axis
- None of these
- Polar axis

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

The point $p(0,\theta)$ in polar coordinate system lies on .....

Answer ( Please select your correct option )

VuAnswers.com

- Polar axis
- y-axis
- Pole
- None of these

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the integral $\iint_R f(x,y)dxdy$, after converting to polar coordinates, it will become . . . . . where $a \le \theta \le b$ and $c \le r \le d$.

Answer ( Please select your correct option )

VuAnswers.com

- $\int_a^b \int_c^d f(r,\theta) r dr d\theta$
- $\int_a^b \int_c^d f(r,\theta) dr d\theta$
- $\int_a^b \int_c^d f(r,\theta) r d\theta dr$
- $\int_a^b \int_c^d f(r,\theta) d\theta dr$

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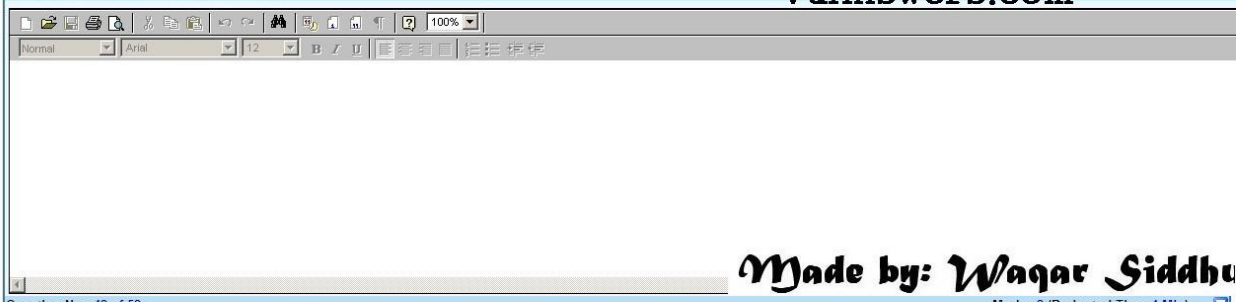
Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Determine whether the following differential is exact or not.

$dz = 4x^3y^3\,dx + 3x^4y^2\,dy$

Answer ( Please click here to Add Answer )

VuAnswers.com



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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Evaluate

$$\int_{-\pi}^{\pi} \cos nx\,dx$$

where n is an integer other than zero.

Answer ( Please click here to Add Answer )

VuAnswers.com



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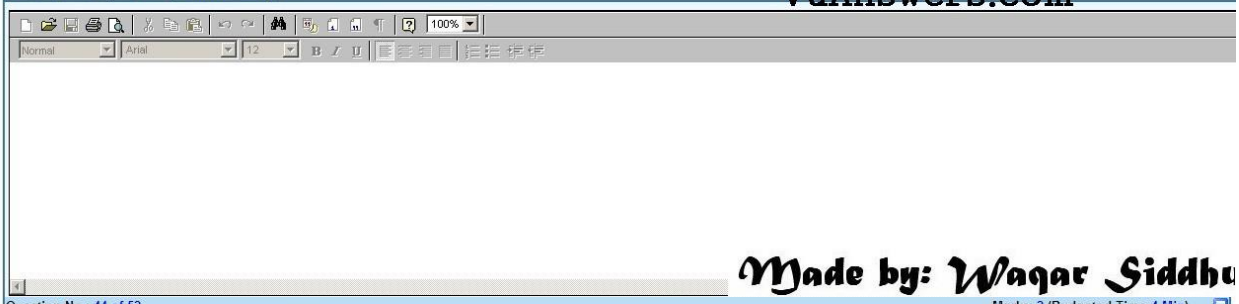
Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Prove whether the following function is even, odd or neither.

$f(x) = x^2 - 4\sin x$

Answer ( Please click here to Add Answer )

VuAnswers.com



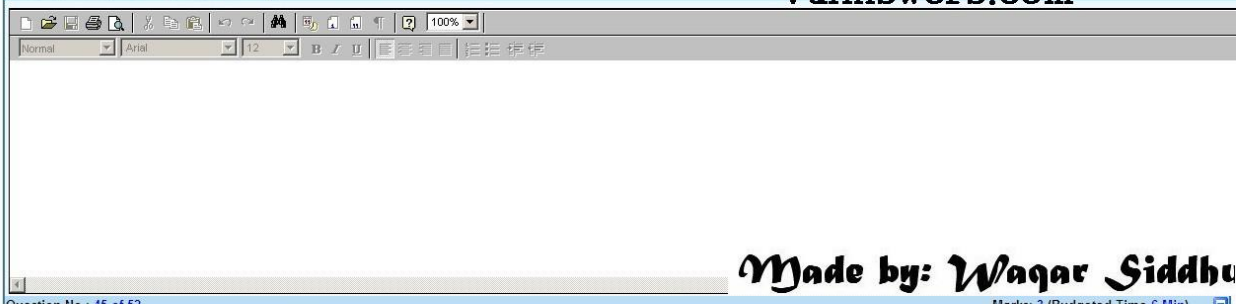
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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Given $\vec{a} \times \vec{b} = 3xi + 2yj + xk$ and $\vec{c} = 7xi + 4yk$. Find scalar triple product of these vectors.

Answer ( Please click here to Add Answer )

VuAnswers.com



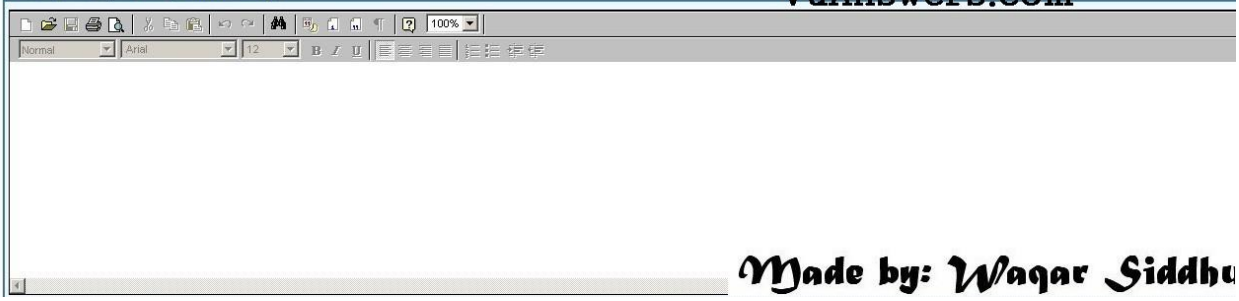
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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = (4+3t)\,\hat{i} + (2-2t)\,\hat{j} + (5+t)\,\hat{k}$ when $3 \le t \le 4$?

Answer ( Please click here to Add Answer )

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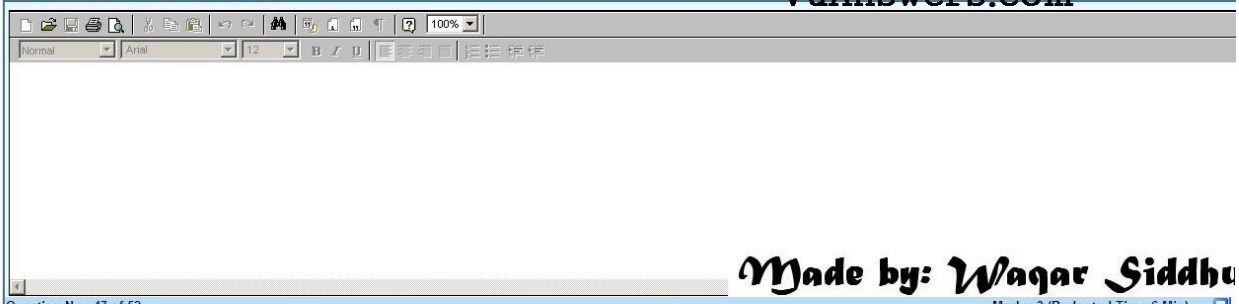
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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Find $div\ \vec{F}$, if $\vec{F} = (3x + y)\hat{i} + xy^2z\,\hat{j} + (xz^2)\hat{k}$

Answer ( Please click here to Add Answer )

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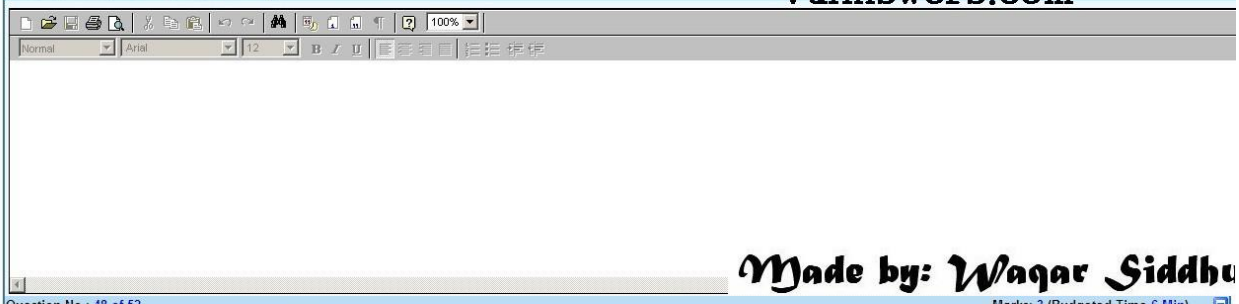
Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Determine the Fourier co-efficient $a_0$ of the periodic function defined below:

$f(x) = 2x + 1 \qquad 0 < x < 2$

Answer ( Please click here to Add Answer )

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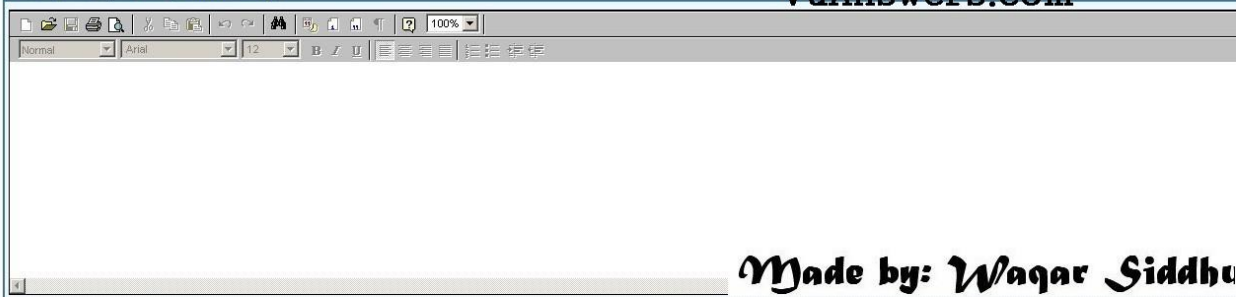
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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

A line, in three dimensional space, passes through the point $(3, -4, 2)$ and parallel to the vector $\vec{r} = 4\hat{i} + 3\hat{j} + 6\hat{k}$. Write down the equation of this line in parametric and symmetric form.

Answer ( Please click here to Add Answer )

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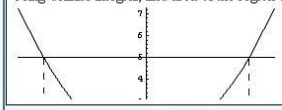
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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Using definite integral, find area of the region bounded by the curves of $y = x^2 + 1$ *and* $y = 5$



Answer ( Please click here to Add Answer )

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Show that Laplace transform of the function

$f(t) = 1$ is

$\frac{1}{s}$ where $s$ is a constant for the integration and $s > 0$.

Answer ( Please click here to Add Answer )

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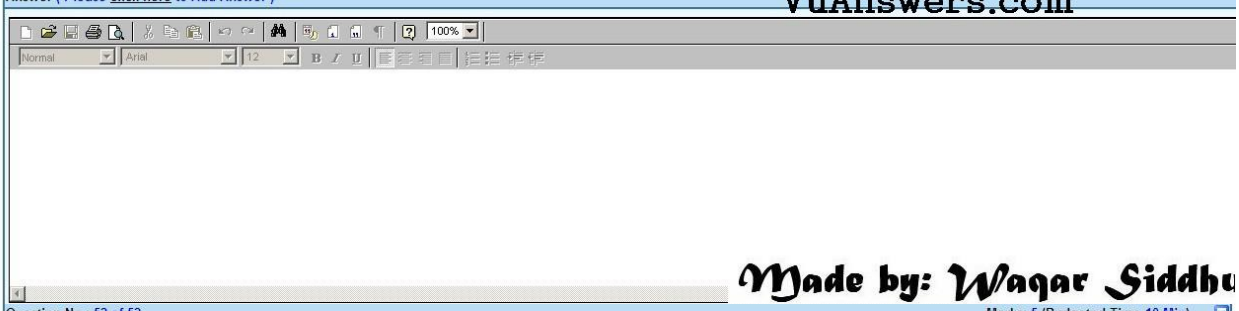
Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$

Answer ( Please click here to Add Answer )

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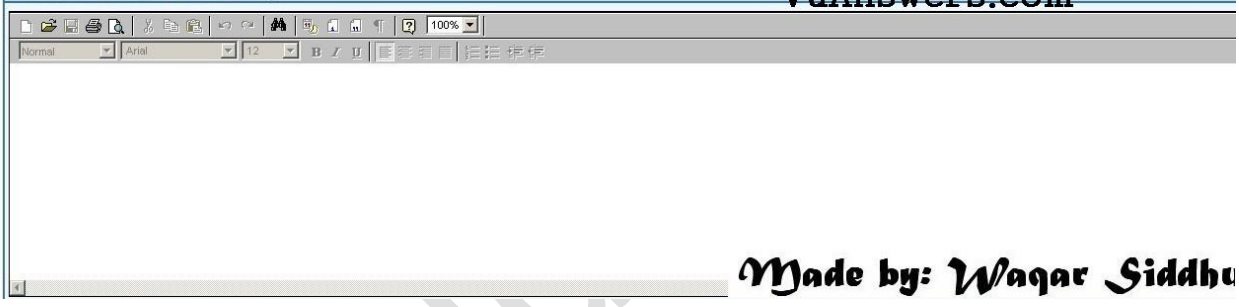
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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider the point $(-5, 5, 6)$ in rectangular coordinate system. Convert it into Spherical coordinates.

Answer ( Please click here to Add Answer )

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